Literature Review

Smart Budget Travel & Fare Optimizer

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Team Members: Yassin Alaa El Din (Lead), Jana Khaled, Peter Fadel, Mohamed Khaled
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1. INTRODUCTION

Modern online travel platforms generate massive volumes of dynamic pricing data across flights, hotels, and ground transportation. Designing an intelligent budget travel optimizer requires combining multi-source web scraping, time-series forecasting, dimensional data warehousing, automated pipeline orchestration, and real-time data quality validation.

2. WEB DATA INGESTION & PIPELINE ORCHESTRATION

Web scraping frameworks (Scrapy, Selenium, BeautifulSoup) enable automated extraction from unstructured web environments [1]. To ensure resilience against frequent structural website changes, modern architectures utilize two-stage staging: storing unparsed HTML/JSON payloads in document stores (MongoDB) before performing structured ETL transformations [2]. Workflow orchestrators like Apache Airflow represent data pipelines as Directed Acyclic Graphs (DAGs), ensuring dependency enforcement and failure logging [3].

3. DIMENSIONAL DATA WAREHOUSING

Kimball's dimensional modeling methodology remains the standard for analytical querying [4]. Structuring travel data into Fact tables (flight bookings, hotel stays) surrounded by Dimension tables (dates, carriers, destinations) avoids multi-table join duplication and accelerates complex budget optimization queries.

4. TIME-SERIES PRICE FORECASTING & OPTIMIZATION

Dynamic airline and hotel fare fluctuations exhibit strong seasonality, lead-time dependence, and day-of-week effects [5]. Time-series methodologies such as Facebook Prophet and gradient-boosted trees (XGBoost) effectively model price trajectories to generate actionable buy vs. wait recommendations [6]. Optimization algorithms combine these predicted fare baselines with user travel constraints to derive minimum-cost package bundles [7].

5. DATA QUALITY ASSURANCE & TRUST LAYER INTEGRATION

Automated data quality frameworks maintain pipeline integrity by evaluating incoming data batches against statistical anomaly baselines and strict schema definitions [8]. Integrating data quality findings directly into user-facing dashboards ensures that downstream analytics and recommendations remain reliable and trustworthy [9].

6. PROJECT CONTRIBUTION

This project bridges the gap between raw web scraping and intelligent decision support by integrating automated ingestion, time-series prediction, package optimization, and a real-time data trust monitoring layer into a unified, user-centric travel platform [10].

7. FORMAL REFERENCES

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